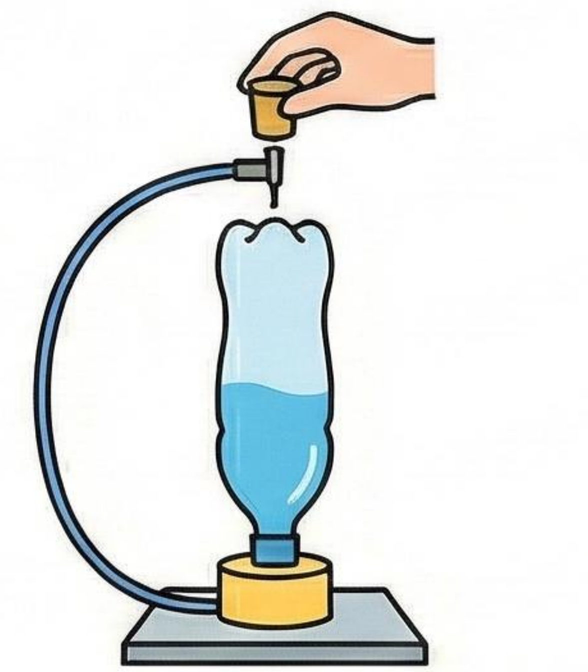
BOTTLE ROCKET SCIENCE EXPERIMENT: LAUNCH METHOD

1. PREPARATION & MATERIALS

Gather items. Wear safety goggles. Fill bottle ~1/3 with water (fuel).

2. ASSEMBLY & FUELING

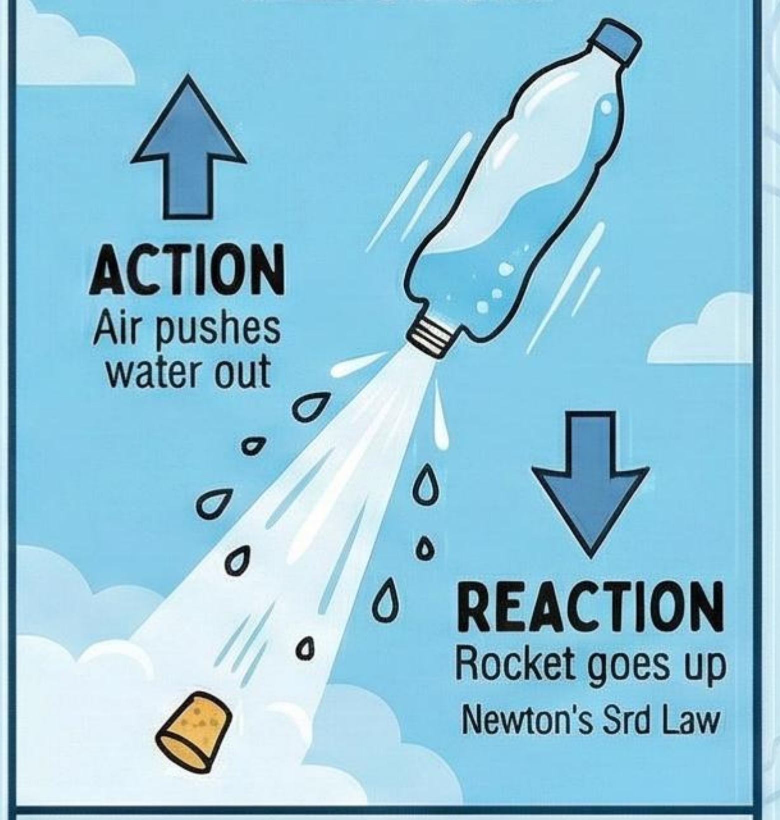


Insert stopper tightly with pump needle. Secure inverted bottle on launch pad.

3. PRESSURIZATION

Pump air into the bottle to build pressure. Keep a safe distance.

4. LAUNCH!



Release stopper (or pull cord). The escaping pressurized water propels the rocket upward!

SCIENCE BITE

Air pressure builds inside. When released, the air forces the water out at high speed, creating thrust (reaction) that launches the bottle (action).

ELEPHANT TOOTHPASTE:
A CATALYTIC CHEMICAL ERUPTION!

MATERIALS NEEDED

Hydrogen Peroxide (H_2O_2)
The oxidizer, source of oxygen.

Dish Soap
Traps oxygen gas to create bubbles.

Food Coloring
Adds colorful stripes.

Yeast (Catalyst)
Contains catalase enzyme, speeds up reaction.

Warm Water
Activates the yeast.

Yeast
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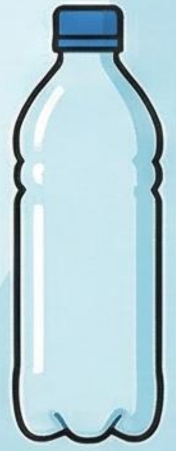


Empty Bottle
Reaction vessel.

THE PROCEDURE

STEP 1: PREPARE THE BASE

Pour Hydrogen Peroxide, Soap, and Food Coloring into the bottle. Gently swirl to mix.

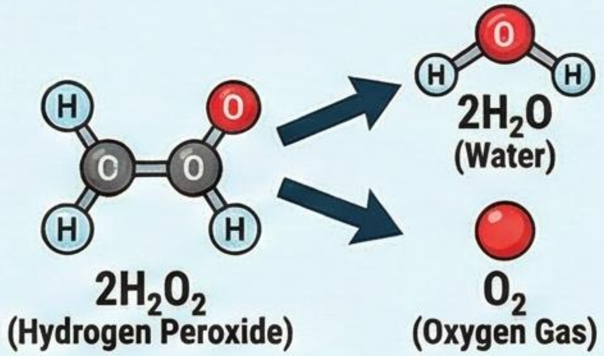
STEP 2: ACTIVATE THE CATALYST

In a separate cup, mix Yeast and Warm Water. Let sit for a minute to activate.

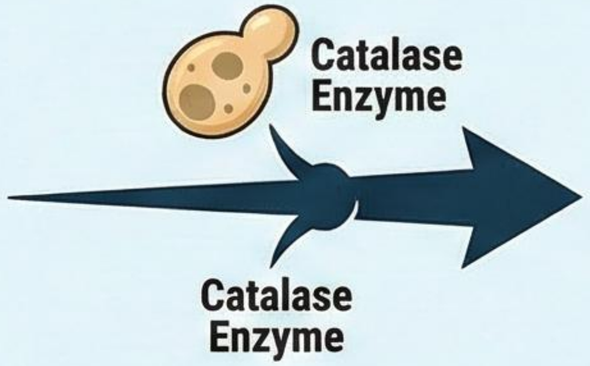
STEP 3: THE ERUPTION!

POUR the yeast mixture into the bottle and **STAND BACK!** Watch the foamy explosion!

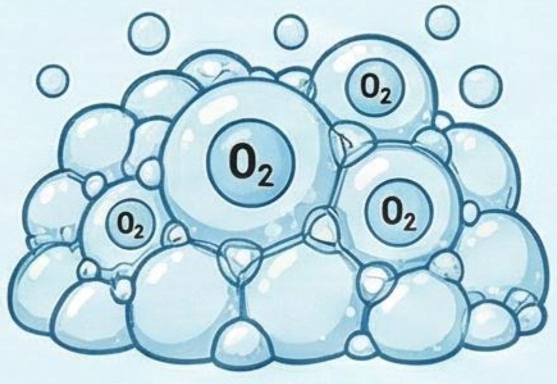
THE SCIENCE BEHIND THE FOAM



CHEMICAL REACTION
Decomposition: Hydrogen Peroxide breaks down into Water and Oxygen Gas.



THE CATALYST EFFECT
Yeast (catalyst) speeds up the decomposition process significantly.



FOAM FORMATION
Soap traps the rapidly released Oxygen Gas, creating a giant volume of foam.

EXOTHERMIC REACTION
Heat Released! The reaction is exothermic, making the bottle and foam warm.

SAFETY FIRST: Always wear safety goggles. Adult supervision required. Do not touch the foam immediately as it may be hot or contain unreacted peroxide.